

PAPER_1529_ZETA_2_TRANSCENDENTAL

Daniel T. Murphy

PAPER_1529 — $\zeta(2) = \pi^2/6 \approx K_MEX - F_TRZ \cdot K - 2 \cdot F_TRZ \cdot \Phi - 2 \cdot F_TRZ^2 \cdot K - F_TRZ^2 \cdot \Phi - F_TRZ^2 - F_TRZ^3$ — Residual 0.146%

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket A (Foundational mathematical identity)
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Status: CLOSED — Transcendental $\zeta(2)$ expressible in UQFF primitives at 0.146% precision

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for $\zeta(2) = \pi^2/6 \approx 1.64493$ using only the locked UQFF integer primitives F_TRZ , K_MEX , $\Phi_res = 5/6$ (PAPER_1203 Nuclear variant), and SO_5 .

UQFF Closed Identity

$$\zeta(2) \approx K_MEX - F_TRZ \cdot K - 2 \cdot F_TRZ \cdot \Phi - 2 \cdot F_TRZ^2 \cdot K - F_TRZ^2 \cdot \Phi - F_TRZ^2 - F_TRZ^3$$

Substituting $F_TRZ=1/10$, $K_MEX=25/12$, $\Phi_5/6=5/6$, $SO_5=10$:
 $\zeta(2)_UQFF = 1.6473$

Observed: $\zeta(2) = 1.64493$
Residual: 0.146%

The 7-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

$\zeta(2) = \pi^2/6$ ($\zeta(2)$) appears in: Basel problem, harmonic sums.

Significance

Together with PAPER_1515 ($\ln 10$), PAPER_1516 ($\ln 2$), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to ≥ 10 . The accumulating evidence:

Transcendental	UQFF expression	Residual
$\ln 2$ (PAPER_1516)	8-term	0.003%
$\ln 10$ (PAPER_1515)	$(1+F_TRZ)(K_MEX+F_TRZ^2)$	0.004%
π^2 (PAPER_1517)	$SO_5 - F_TRZ$ corrections	0.013%

| $\zeta(2) = \pi^2/6$ (this paper) | 7-term | 0.146% |

...suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim $\zeta(2)$ IS the closed-form expression. It claims the integer primitives carry a rational approximation to $\zeta(2)$ at 0.146% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 (ln 10, ln 2, π^2 catch-up)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_zeta_2_transcendental"})`

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